

Deep Learning Innovation Challenge | Project Plan: Automating Smart Money Concepts on SPY

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Part 1: Think Big

What is the innovative idea?

Smart Money Concepts (SMC) is a price-action framework, largely derived from the Inner Circle Trader (ICT) methodology, used to identify structural footprints that institutional participants leave in candlestick data. The premise is that large players must move price to specific zones to fill orders, creating repeatable, detectable structures: Fair Value Gaps (FVGs), Order Blocks (OBs), Break of Structure (BOS), Change of Character (CHoCH), Liquidity Pools, and Supply and Demand Zones. Today every SMC practitioner draws these structures by hand, making the methodology entirely subjective and impossible to backtest at scale.

This project builds a deep learning system that detects these structures automatically from raw OHLCV candlestick data on SPY (SPDR S&P 500 ETF). The output is a set of zone labels with confidence scores, visualised as interactive chart overlays. This is not a price prediction project. The model does not forecast whether SPY will rise or fall; it identifies which structural zones are present and where, enabling objective, reproducible analysis of institutional market structure.

What problem does it address?

SMC has no standardised algorithmic implementation. Rule-based scripts approximate the textbook definitions but are context-blind: a fair value gap formed during high-volume post-BOS expansion carries different structural weight than one formed during a low-volume drift, yet a fixed rule treats them identically. A trained deep learning model learns those contextual relationships. Beyond detection, the project makes systematic backtesting of SMC hypotheses feasible at scale and opens the door to objective confluence scoring, quantifying how many independent structures align at a given price level, something currently computed only by intuition.

Which data will be used?

SPY hourly OHLCV data from 2018 to present via yfinance (free, no API key required, approximately 16,000 candles). The hourly timeframe is the practitioner consensus for SMC zone detection: daily candles provide too few samples for effective deep learning training, while sub-15-minute candles introduce noise and labelling ambiguity (Mind Math Money, 2026; Blueberry Markets, 2025). A well-documented SMC principle is that structures carry more significance when they align across timeframes: the 1-hour chart identifies zones while the daily chart provides macro bias. This multi-timeframe dependency maps architecturally to a temporal feature fusion design (Lim et al., 2021) and will be introduced as an extension once the single-timeframe baseline is validated.

Training labels are generated programmatically via the smart-money-concepts Python library (Attridge, 2023), which implements deterministic rule-based algorithms for FVG, OB, BOS, and liquidity detection. This approach is known as weak supervision or programmatic labelling, the methodology behind Google's Snorkel framework (Ratner et al., 2020). No manual annotation is required. The deep learning contribution is not the labelling: the model learns which contextual features predict whether a zone will behave as theory describes, a generalisation the rules alone cannot provide.

Why is this interesting and innovative?

The academic literature on financial deep learning is dominated by price direction prediction and classical indicator recognition. Fischer and Krauss (2018) showed that LSTM-based excess returns on S&P 500 constituents largely disappeared after 2010 as the approach was widely adopted. SMC has received virtually no academic treatment and no published deep learning application. Detecting SMC structures is a sequence labelling problem requiring temporal context across hundreds of candles, which is exactly the problem class where recurrent and attention-based architectures outperform classical methods. Chen and Tsai (2020) achieved 90.7% accuracy on candlestick pattern classification with a CNN; Ramadhan et al. (2022) reported 82.7% on CNN-LSTM models. This project extends that line of work into the more complex, context-dependent domain of institutional market structure analysis.

Are there ethical aspects involved?

Market manipulation: automating liquidity pool detection could, at institutional scale, be used to engineer the stop hunts it detects. The project is strictly for research and educational purposes. **Data bias:** historical training risks regime overfitting; a strict temporal train/test split and honest documentation of limitations will address this. **Retail harm:** results will be communicated with uncertainty estimates and no trading edge will be claimed.

Part 2: The First Step

What is the realistic first step?

Build a complete end-to-end pipeline: automated data retrieval, programmatic FVG labelling, a trained baseline LSTM, and an interactive Plotly visualisation comparing model-detected and rule-generated zones on real SPY data. Scoping to FVG detection only is deliberate: FVGs have the clearest algorithmic definition, the most consistent labels, and are immediately verifiable on a chart without deep domain knowledge.

- **Data:** download SPY H1 OHLCV 2018-2024 via yfinance, remove zero-volume candles, apply rolling min-max normalisation.
- **Labels:** apply smart-money-concepts to generate per-candle ternary FVG labels (bullish / bearish / none).
- **Model:** format as 60-candle sliding windows (60x5 tensors), train a two-layer stacked LSTM with dropout and weighted cross-entropy loss to handle class imbalance.
- **Visualisation:** Plotly candlestick chart with rule-generated and model-predicted FVG zones rendered side by side.

Which techniques and tools will be used?

The project uses a staged model comparison rather than assuming any single architecture is optimal.

- **Stage 1 (LSTM):** two-layer stacked LSTM, 64 units per layer with dropout. Standard sequential baseline for financial time series.
- **Stage 2 (CNN-LSTM):** 1D convolutional front-end (kernel size 3-5) for local formation extraction, followed by LSTM for temporal context. FVGs are inherently three-candle local formations, making a convolutional front-end well-motivated. Literature reports 82-91% accuracy on comparable candlestick tasks (Ramadhan et al., 2022; Chen and Tsai, 2020).
- **Stage 3 (Transformer, time-permitting):** lightweight encoder with multi-head self-attention attending across the full window simultaneously. The Temporal Fusion Transformer (Lim et al., 2021) is a candidate if the project extends to multi-timeframe inputs. Whether this outperforms CNN-LSTM is an open empirical question.

Known bottlenecks: (1) Class imbalance: FVGs present in ~5-10% of candles; F1 on the positive class is the primary metric, not accuracy. (2) Lookahead leakage: some labels are confirmed only after the fact; the pipeline must encode only information available at the time of the candle. (3) Label noise: rule-based labels approximate expert judgement and set a natural ceiling on performance. (4) Non-stationarity: rolling normalisation partially mitigates regime drift but does not eliminate it.

Tools: Python, PyTorch, yfinance, pandas, numpy, smart-money-concepts, Plotly, scikit-learn.

When is the first step successful?

- **Pipeline completeness:** data download, labelling, training, and chart rendering all execute from a single script without manual intervention.
- **Model performance:** the LSTM achieves a positive F1 score on the FVG minority class in the 2023-2024 holdout set, meaningfully above the naive majority-class baseline.
- **Qualitative output:** the Plotly chart displays at least one concrete case where the model's prediction differs from the rule engine, with a plausible explanation for the discrepancy.

No specific F1 threshold is committed to at this stage. This task has no prior benchmark in the literature, and a numerical target set before any baseline run would be methodologically unsound. A result above the naive baseline with honest qualitative analysis constitutes a credible first milestone.

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